

Hello

I'm Rana Sandhu

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M.Sc. in Computational Neuroscience and Cognitive Robotics and an interdisciplinary background.
Experience analyzing and visualizing data, programming, and research work.

Skills

- Python with experience utilizing packages such as Flask, Matplotlib, Seaborn, SciPy, NumPy, pandas, TensorFlow, Keras, and PyTorch
- MATLAB experience with image processing toolbox and programming neuronal models
- Working in Linux, Git and GitHub, basic JavaScript
- Language Skills: English (native); Mandarin (proficient); Spanish (proficient); Hindi (proficient)

Education

OCTOBER 2018 - DECEMBER 2019

University of Birmingham, Birmingham UK - M.Sc. in Computational Neuroscience and Cognitive Robotics

- Collaborated with group to program a Pioneer 3-DX (robot) to patrol a known area and implemented object recognition to differentiate trash and recycling
- Worked with sensor data, and implemented a Monte Carlo filter, for localization of robot
- Created a neural network to determine orientation of an object in images
- Developed visuomotor experiment, in python, where participants used either a joystick or an ATI 6-axis force sensor to control a cursor on screen
- Analyzed resulting data from experiment for trends and statistical significance

SEPTEMBER 2011 - MAY 2016

University of Maryland, College Park - B.Sc. in Physiology and Neurobiology, Minor in Chinese Language

Experience

SEPTEMBER 2020 - PRESENT

TalkitOver, Remote, UK Based - Data Scientist

- Work with group using Git to help build and maintain helpline chatbot
- Data analysis, and data visualization of chat data, using Python, Pandas, Matplotlib, and Seaborn
- Maintain website, built with Flask, JavaScript, HTML, and CSS
- Implemented Sentence Encoder to vectorize messages for classification
- Improved chatbot algorithm, leading to improved happiness (self-reported by user)
- Conduct weekly meetings to discuss improvements for the chatbot based on data analysis
- Create documentation for the chatbot code
- Test machine learning techniques/pre-trained models for future implementation

JUNE 2016 - APRIL 2018

Hackensack University Medical Center, Hackensack NJ - Research Assistant

- Worked with cancer tissue slides and cultured cancer cells for various experiments
- Grow and monitor cells used in the experiments
- Conducted experiments such as: PCR, Western Blot, RNA extraction
- Worked with team to optimize protocols for experiments
- Interpreted and quantified results, and presented them weekly

Volunteer

MAY 2021 - PRESENT

AI for Good, Remote, UK Based - Data Analyst

- Clean data and look for relationships
- Test different NLP models to determine climate misinformation

APRIL 2022 - PRESENT

Child Mind Institute, Remote, New York - Data Analyst

- Image analysis: create functions to determine the difference between user input and given image, then extract features